George Amgad

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Education

German University in Cairo (GUC) Expected Graduation: June 2027
Bachelor of Science in Computer Science (final year) *Cairo, Egypt*

- GPA: 1.05 (German scale, lower is better); planning to continue to a Master's degree after graduation.
- Academic scholarship for top 1% national IGCSE performance (124% overall, 2022).
- Coursework: Operating Systems, Computer Architecture, Embedded Systems, Data Engineering, Computer Vision.

Experience

Hiryo | React Native, Expo, Supabase Jun 2025 – Feb 2026
Mobile Software Engineer: Intern (Jun – Aug 2025), then Part-Time (Aug 2025 – Feb 2026) *Hybrid*

- Built background location sharing, geofencing, shared maps, and SOS alerts for a family safety app, resolving Android constraints around permissions, background execution, and battery optimization.
- Simplified onboarding and permission flows to reduce early user drop-off, and managed Google Play releases including internal testing tracks and compliance.
- Built a household management app with shared tasks, events, and document storage, using Supabase for real-time synchronization and notifications.

KemitWare Aug 2024 – Dec 2024
Software Engineer (Intern) *Remote*

- Developed cross-platform React Native apps for Android and iOS, integrating authentication and real-time messaging with backend APIs.

Nefertari International Schools Jul 2021 – Jun 2022
Teaching Assistant, A-Level Mathematics (Paid) *Cairo, Egypt*

- Supported teaching of A-Level Mathematics for 50 students, explaining problem-solving methods, created and graded quizzes and helped with exam preparation.

Projects

Bachelor Thesis: Tablet Builder Puzzle | Web-based educational game 2026

- Built a game teaching K–12 students how tablet hardware works, with Bloom's Taxonomy-aligned difficulty tiers; field-tested with 21 students across three age groups, showing a statistically significant learning gain (Wilcoxon signed-rank test).

Natural Language to SQL Interface | Python, Gemini API, MS SQL Server 2026

- Built a system that converts natural-language questions into valid, executable SQL queries, with safety guardrails on generated queries.

F1 Podium Prediction | Python, pandas, scikit-learn, Keras 2026

- Predicted podium finishes from post-qualifying data (1950–2024) with a tree ensemble and a neural network on time-based splits, beating the baseline on PR-AUC.

Vehicle Detection, Tracking & Counting | Python, YOLO, ByteTrack 2026

- Building a team pipeline that detects, tracks, and counts vehicles in UA-DETRAC traffic video.

Stock Market Data Engineering Pipeline (with Finaria) | Python, Docker 2026

- Cleaning Finaria stock data in a Dockerized pipeline to test its realism for bank crisis simulations.

ReadFlow | React Native, Expo 2026

- Built a working cross-platform mobile reading app with a minimal, distraction-free reader interface.

Systems Projects | C, C++

- Tomasulo out-of-order execution simulator (C++), OS process management simulator (C), embedded color sorter (C).

Events Management System | React, Node.js, Express, MongoDB 2025

- University events platform built by a 10-person Agile team; worked on backend logic, schema design, and integration.

Technical Skills

Languages: Python, C++, C, Java, JavaScript, TypeScript, SQL
AI/ML & Data: pandas, scikit-learn, Keras, YOLO, Gemini API, MS SQL Server, MongoDB, Firebase, Supabase
Mobile, Web & Tools: React Native, Expo, React, Node.js, Express, Git, Docker, Vercel, Figma

Leadership & Activities

IEEE GUC | Team Support Member 2026 – Present

- Coordinate across club teams, resolve conflicts, and organize trips and outings for members.

Basketball Player, U21 Cairo League | Heliopolis Sporting Club